Decimal Operations

USA Grade 6 Math
CCSS 6.NS.B.3

Name: ______________________________ Date: __________________

LEARNING TARGET

I can fluently compute with multi-digit decimals.

KEY STRATEGY + WORKED EXAMPLE

Estimate, compute using whole-number patterns, place the decimal by value, and verify using inverse operations.

18.6 / 3 = 6.2; check 6.2 x 3 = 18.6.

A. TRY TOGETHER - USE THE STRATEGY

1. 18.75 / 2.5 = ___
2. 4.68 * 3.2 = ___
3. 96.6 / 4.2 = ___
4. 7.05 * 6.4 = ___

B. CORE PRACTICE - SHOW YOUR WORK

5. 125.4 / 3.8 = ___
6. 0.875 * 2.4 = ___
7. 54.72 / 1.8 = ___
8. 12.06 * 4.5 = ___
9. 83.16 / 2.7 = ___
10. 9.375 * 1.6 = ___
11. 18.75 / 2.5 = ___
12. 4.68 * 3.2 = ___

C. INDEPENDENT FLUENCY - CALCULATE AND CHECK

13. 96.6 / 4.2 = ___
14. 7.05 * 6.4 = ___
15. 125.4 / 3.8 = ___
16. 0.875 * 2.4 = ___
17. 54.72 / 1.8 = ___
18. 12.06 * 4.5 = ___
19. 83.16 / 2.7 = ___
20. 9.375 * 1.6 = ___

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D. APPLY THE SKILL

21. A 7.5-kg bag is split into 12 equal portions. Find each mass.

22. Fabric costs $8.75 per yard. Find cost of 3.6 yards.

23. A car uses 12.8 gallons for 358.4 miles. Find miles per gallon.

E. REASON, CHECK AND CORRECT

24. Explain why counting decimal places is useful for multiplication but not a complete division strategy.

25. Correct: 6.4 x 0.5 = 32.

EXIT TICKET + CHALLENGE

26. Solve 45.36 / 1.2 and verify by multiplication.

Challenge: Write a two-step money problem using decimal multiplication and subtraction.

F. STUDENT REVIEW - SHOW WHAT YOU UNDERSTAND

Use your own words so your teacher can see what you understand and what to practise next.

27. Explain one strategy you used today.

28. Check one answer in a different way.

29. Circle one: I understand it | I need more practice | I need help

30. The easiest part was: ____

31. The most difficult part was: ____

Why was it difficult? ____

32. My next practice step is: ____

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TEACHER COPY - ANSWERS AND SUPPORT

ANSWER KEY

A. 1) 7.5, 2) 14.976, 3) 23, 4) 45.12

B. 5) 33, 6) 2.1, 7) 30.4, 8) 54.27, 9) 30.8, 10) 15, 11) 7.5, 12) 14.976

C. 13) 23, 14) 45.12, 15) 33, 16) 2.1, 17) 30.4, 18) 54.27, 19) 30.8, 20) 15

D. 21) 0.625 kg. 22) $31.50. 23) 28 mpg.

E. 24) Division needs scaling divisor and dividend equally; estimation confirms placement. 25) 6.4x0.5=3.2.

EXIT. 26) 37.8.

Challenge: Answers vary. Check that the student's example follows every condition and that the reasoning is mathematically correct.

F. 27-28) Answers vary. Look for a clear strategy and a valid second check. 29-32) Use the reflection to identify confidence, difficulty and the next teaching step.

COMMON MISCONCEPTIONS

Students may place decimals by appearance rather than magnitude. Estimate first and use an inverse operation to check.

TEACHER CHECK

[ ] Uses an appropriate Grade 6 Math strategy.

[ ] Shows or explains thinking clearly.

[ ] Checks a response using evidence, a rule, or a second method.

[ ] Identifies a specific difficulty and useful next practice step.

PARENT / TEACHER TIP

Ask the child to explain how they know, not only state an answer. If the explanation is unclear, use small objects, a drawing, or a number line before returning to symbols.

NEXT STEP

Revisit the item the child marked difficult. Model one example, solve one together, then ask the child to complete one independently and explain the strategy.